Assignment_3

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2. Problem to demonstrate the role of qualitative (nominal) predictors in addition to quantitative predictors in multiple linear regression

Attach “Credits” data from R. Regress “balance” on (a) “gender” only.

```
data("Credit", package = "ISLR")
model21 = lm(Balance~Gender, data = Credit)
summary(model21)

##
## Call:
## lm(formula = Balance ~ Gender, data = Credit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -529.54 -455.35  -60.17  334.71 1489.20
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    509.80      33.13  15.389   <2e-16 ***
## GenderFemale    19.73      46.05   0.429    0.669
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 460.2 on 398 degrees of freedom
## Multiple R-squared:  0.0004611,  Adjusted R-squared:  -0.00205
## F-statistic: 0.1836 on 1 and 398 DF,  p-value: 0.6685
```

(b) “gender” and “ethnicity” .

```
model22 = lm(Balance~Gender+Ethnicity, data = Credit)
summary(model22)

##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity, data = Credit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -540.92 -453.61  -56.37  336.24 1490.77
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)          520.88      51.90  10.036   <2e-16 ***
```

```
## GenderFemale          20.04      46.18   0.434    0.665
## EthnicityAsian       -19.37      65.11  -0.298    0.766
## EthnicityCaucasian   -12.65      56.74  -0.223    0.824
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 461.3 on 396 degrees of freedom
## Multiple R-squared:  0.000694,   Adjusted R-squared:  -0.006877
## F-statistic: 0.09167 on 3 and 396 DF,  p-value: 0.9646
```

```
model22_interaction = lm(Balance~Gender+Ethnicity+(Gender*Ethnicity), data =
Credit)
summary(model22_interaction)
```

```
##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity + (Gender * Ethnicity),
##     data = Credit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -563.11 -445.84  -57.75  333.87 1483.66
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      553.45      65.95   8.392 8.65e-16 ***
## GenderFemale                     -44.45      92.80  -0.479    0.632
## EthnicityAsian                  -100.58      94.25  -1.067    0.287
## EthnicityCaucasian               -38.11      80.91  -0.471    0.638
## GenderFemale:EthnicityAsian      154.69     130.46   1.186    0.236
## GenderFemale:EthnicityCaucasian   50.61     113.57   0.446    0.656
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 461.6 on 394 degrees of freedom
## Multiple R-squared:  0.004472,   Adjusted R-squared:  -0.008161
## F-statistic: 0.354 on 5 and 394 DF,  p-value: 0.8796
```

(c) “gender”, “ethnicity”, “income”.

```
model23 = lm(Balance~Gender+Ethnicity+Income, data = Credit)
summary(model23)
```

```
##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity + Income, data = Credit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -794.14 -351.67  -52.02  328.02 1110.09
##
```

```
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        230.0291    53.8574   4.271 2.44e-05 ***
## GenderFemale        24.3396    40.9630   0.594    0.553
## EthnicityAsian       1.6372    57.7867   0.028    0.977
## EthnicityCaucasian   6.4469    50.3634   0.128    0.898
## Income               6.0542     0.5818  10.406  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 409.2 on 395 degrees of freedom
## Multiple R-squared:  0.2157, Adjusted R-squared:  0.2078
## F-statistic: 27.16 on 4 and 395 DF,  p-value: < 2.2e-16
```

```
model23_interaction =
lm(Balance~Gender+Ethnicity+(Gender*Ethnicity)+(Ethnicity*Income)+(Income*Gen
der), data = Credit)
summary(model23_interaction)
```

```
##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity + (Gender * Ethnicity) +
##     (Ethnicity * Income) + (Income * Gender), data = Credit)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -761.0 -350.3  -52.8  333.7 1104.2
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     177.3633    84.0373   2.111   0.0354 *
## GenderFemale                     -3.9633   100.5685  -0.039   0.9686
## EthnicityAsian                   15.5548   108.5243   0.143   0.8861
## EthnicityCaucasian               94.1012    98.1493   0.959   0.3383
## Income                            7.7178     1.2375   6.236 1.17e-09 ***
## GenderFemale:EthnicityAsian      81.0004   116.4601   0.696   0.4871
## GenderFemale:EthnicityCaucasian  57.7073   101.0052   0.571   0.5681
## EthnicityAsian:Income            -1.1207     1.5876  -0.706   0.4807
## EthnicityCaucasian:Income        -2.5335     1.3902  -1.822   0.0691 .
## GenderFemale:Income              -0.5246     1.2083  -0.434   0.6644
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 409.7 on 390 degrees of freedom
## Multiple R-squared:  0.2238, Adjusted R-squared:  0.2059
## F-statistic:  12.5 on 9 and 390 DF,  p-value: < 2.2e-16
```

(d) Output all the regressions in (a)-(c) in a single table using stargazer. Comment on the significant coefficients in each of the models.

```
library(stargazer)
```

```
## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
Statistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
stargazer(model21, model22, model23, type = "text", title = "Regression
Results for Balance", out = "balance_regression_results.txt")
```

```
##
## Regression Results for Balance
##
===============================================================================
======
##                                             Dependent variable:
##                     ---------------------------------------------------------
---------
##                                                   Balance
##                             (1)                   (2)                    (3)
## -----------------------------------------------------------------------------
---------
## GenderFemale              19.733                20.038                 24.340
##                          (46.051)              (46.178)
(40.963)
##
## EthnicityAsian                                  -19.371                 1.637
##                                                (65.107)
(57.787)
##
## EthnicityCaucasian                              -12.653                 6.447
##                                                (56.740)
(50.363)
##
## Income
6.054***
##
(0.582)
##
## Constant                 509.803***            520.880***
230.029***
##                           (33.128)              (51.901)
(53.857)
##
## -----------------------------------------------------------------------------
---------
## Observations                400                   400                    400
## R2                         0.0005                0.001                  0.216
```

```
## Adjusted R2                -0.002               -0.007                  0.208
## Residual Std. Error 460.230 (df = 398)  461.337 (df = 396)    409.218 (df
= 395)
## F Statistic         0.184 (df = 1; 398) 0.092 (df = 3; 396) 27.161*** (df
= 4; 395)
##
================================================================================
======
## Note:                                                    *p<0.1; **p<0.05;
***p<0.01
```

```
stargazer(model21, model22_interaction, model23_interaction, type = "text",
title = "Regression Results for Balance Interaction", out =
"balance_regression_interaction_results.txt")
```

```
##
## Regression Results for Balance Interaction
##
================================================================================
==================
##                                                        Dependent variable:
##                                 ----------------------------------------
---------------------
##                                                            Balance
##                                         (1)                 (2)
(3)
## --------------------------------------------------------------------------
---------------------
## GenderFemale                          19.733              -44.449
-3.963
##                                      (46.051)            (92.796)
(100.569)
##
## EthnicityAsian                                           -100.577
15.555
##                                                          (94.251)
(108.524)
##
## EthnicityCaucasian                                        -38.109
94.101
##                                                          (80.907)
(98.149)
##
## Income
7.718***
##
(1.238)
##
## GenderFemale:EthnicityAsian                               154.686
81.000
```

```
##                                                                  (130.460)
(116.460)
##
## GenderFemale:EthnicityCaucasian                                   50.609
57.707
##                                                                  (113.566)
(101.005)
##
## EthnicityAsian:Income
-1.121
##
(1.588)
##
## EthnicityCaucasian:Income
-2.534*
##
(1.390)
##
## GenderFemale:Income
-0.525
##
(1.208)
##
## Constant                                     509.803***          553.449***
177.363**
##                                               (33.128)            (65.947)
(84.037)
##
## ---------------------------------------------------------------------------
---------------------
## Observations                                    400                 400
400
## R2                                             0.0005               0.004
0.224
## Adjusted R2                                   -0.002              -0.008
0.206
## Residual Std. Error                  460.230 (df = 398)  461.631 (df = 394)
409.699 (df = 390)
## F Statistic                          0.184 (df = 1; 398) 0.354 (df = 5; 394)
12.496*** (df = 9; 390)
##
===========================================================================
==================
## Note:
*p<0.1; **p<0.05; ***p<0.01
```

(e) Explain how gender affects “balance” in each of the models (a)- (c) .

ANSWER: In model for only gender, the R^2 is 0.0004611. Therefore, Gender alone is not a good predictor for Balance. In model for gender and ethnicity, the R^2 is 0.001. Therefore,

R^2 is slightly higher than the previous model but it doe not explain the variance much better either. Hence, this model is also not very good, even though it is better than the previous one. In the model for Gender, Ethnicity and Income, the R^2 is 0.22. Hence, there is a significant portion of the variance in Balance compared to the previous two models. Therefore, it is a much better model than the previous two models.

(f) Compare the average credit card balance of a male African with a male Caucasian on the basis of model (b).

```
coef(model22)["EthnicityCaucasian"]

## EthnicityCaucasian
##          -12.65305
```

The coefficient of EthnicityCaucasian represents the expected difference in average credit card balance between a Caucasian male and an African American male.A negative value indicates African Americans carry higher balances.

(g) Compare the average credit card balance of a male African with a male Caucasian when each earns 100,000 dollars. For comparison, use the model in (c).

```
coef(model23)["EthnicityCaucasian"]

## EthnicityCaucasian
##           6.446938
```

When income is fixed at $100,000, a Caucasian male is expected to have an average credit card balance approximately 6.45 units higher than an African American male. Since income is held constant, this difference represents the independent effect of ethnicity on credit card balance.

(h) Compare and comment on the answers in (f) and (g)

The estimated difference in credit card balance changes from −12.53 in model (b) to +6.45 in model (c) after including income in the regression. This sign reversal indicates that income was a confounding variable. In the model without income, African American males appear to have higher balances because they tend to have lower income levels on average. After controlling for income, the true relationship is revealed: Caucasian males have slightly higher balances. Hence, failing to include income leads to a misleading conclusion.

(i) Based on the model in (c), predict the credit card balance of a female Asian whose income is 2000,000 dollars.

```
new_person <- data.frame(
  Gender = "Female",
  Ethnicity = "Asian",
  Income = 200
)

predict(model23, newdata = new_person)
```

```
##        1
## 1466.851
```

(j) Check the goodness of fit of the different models in (a) -(c) in terms of AIC, BIC and adjusted R2. Which model would you prefer?

```
AIC(model21, model22, model23)

##         df      AIC
## model21  3 6044.527
## model22  5 6048.434
## model23  6 5953.518

BIC(model21, model22, model23)

##         df      BIC
## model21  3 6056.501
## model22  5 6068.391
## model23  6 5977.466

summary(model21)$adj.r.squared

## [1] -0.002050271

summary(model22)$adj.r.squared

## [1] -0.006876514

summary(model23)$adj.r.squared

## [1] 0.207774
```

The model with the highest R^2 is preferred,i.e.,Model (C).

4. Problem to demonstrate the impact of ignoring interaction term in multiple linear regression

Consider a simulation setting where the data is generated as follows:

```
set.seed(123)

n <- 100
R <- 1000

config1 <- c(-2.5, 1.2, 2.3, 0.001)
config2 <- c(-2.5, 1.2, 2.3, 3.1)


run_simulation <- function(n, R, beta_params) {
  beta0 <- beta_params[1]
  beta1 <- beta_params[2]
  beta2 <- beta_params[3]
```

```r
  beta3 <- beta_params[4]

  mse_correct <- numeric(R)
  mse_naive <- numeric(R)

  for (r in 1:R) {
    # Step 1: Generate x1 from Normal(0,1)
    x1 <- rnorm(n, mean = 0, sd = 1)

    # Step 2: Generate x2 from Bernoulli(0.3)
    x2 <- rbinom(n, size = 1, prob = 0.3)

    # Step 3: Generate response with interaction
    epsilon <- rnorm(n, mean = 0, sd = 1)
    y <- beta0 + beta1*x1 + beta2*x2 + beta3*(x1*x2) + epsilon

    # Step 4: Fit both models
    model_correct <- lm(y ~ x1 + x2 + x1:x2)
    model_naive <- lm(y ~ x1 + x2)

    # Computing MSE
    mse_correct[r] <- mean(residuals(model_correct)^2)
    mse_naive[r] <- mean(residuals(model_naive)^2)
  }

  return(list(
    avg_mse_correct = mean(mse_correct),
    avg_mse_naive = mean(mse_naive),
    mse_correct = mse_correct,
    mse_naive = mse_naive
  ))
}

run_simulation(100,1000,c(-2.5,1.2,2.3,0.001))
```

```
## $avg_mse_correct
## [1] 0.9631944
##
## $avg_mse_naive
## [1] 0.9739083
##
## $mse_correct
##    [1] 0.8511631 0.9602055 1.1195283 1.0043441 0.9340622 0.9593344
1.1642569
##    [8] 1.1217837 0.7978635 0.9382235 0.9265837 1.1240546 0.9878345
1.0495433
##   [15] 0.6668817 1.1806558 0.6025595 0.9339389 0.8400421 1.2028140
0.8633813
##   [22] 1.0396428 0.9655059 1.1248303 1.0716287 0.8190131 0.9240596
```

```
0.8843535
##   [29] 0.6744280 0.9594065 1.1208650 1.1101941 0.9813772 1.2152708
0.8430911
##   [36] 1.2270940 1.0253857 1.1018195 0.8591747 0.9338904 0.7121827
0.9649639
##   [43] 1.0669452 1.2806620 0.9574646 1.0840626 1.0005940 0.8219106
1.1961841
##   [50] 0.9380908 1.0340598 1.0619782 1.0055131 0.9719751 0.9622186
0.8783047
##   [57] 0.8798710 0.9631876 0.9282966 0.9892506 0.8134892 0.9759858
0.8908095
##   [64] 0.9958596 0.9997398 0.8272421 0.8642148 1.2440663 1.0556850
0.9838358
##   [71] 0.8523210 0.9755285 0.9104178 0.9629931 0.8014365 0.8772507
0.8260519
##   [78] 1.0479337 0.8041048 1.0284109 0.8043705 1.2396617 0.9307682
0.8792202
##   [85] 1.0052468 0.9068418 0.6728680 0.9841076 0.9301163 0.9200855
1.1005434
##   [92] 1.0220627 0.8038798 0.9072987 0.8796999 1.0559835 0.8851410
1.0213916
##   [99] 1.1977000 1.0053808 0.6830194 1.0359013 1.1257250 1.0222068
1.0414583
##  [106] 0.7133538 0.8980224 0.9385607 1.0619702 0.9841011 0.8098888
0.8077855
##  [113] 0.8377663 1.0108942 1.3325774 1.0718247 0.9465029 0.9248388
0.8482812
##  [120] 0.9475343 0.9357535 0.9616382 1.2739665 0.7247223 0.9643491
0.8197339
##  [127] 1.0164664 0.9169040 0.7566221 0.7616770 0.8755224 1.1988702
1.1218217
##  [134] 0.9436985 1.0839092 0.9210328 0.8733366 0.9115994 1.0970721
1.3838465
##  [141] 0.9299031 0.9842831 1.0557059 0.7613090 0.9428698 0.8144222
1.3202055
##  [148] 0.7998735 1.0462277 1.0323445 1.1670110 0.8373549 0.9644205
0.9748160
##  [155] 1.1487764 0.7440055 0.7344456 1.1146372 1.0718142 1.0083167
0.8633858
##  [162] 0.8546137 0.8302891 0.9910272 1.1725070 0.8044684 1.0692872
1.1052489
##  [169] 1.1927949 0.9967579 0.9502807 0.9110405 0.8373373 0.8212196
1.0750134
##  [176] 1.1156961 0.9118338 0.8684574 1.1811836 1.0319844 0.8362548
0.9166377
##  [183] 1.0031340 0.9571470 0.9113169 0.8251637 1.0658329 0.9817381
0.8999842
##  [190] 0.9082262 0.8690223 1.0239034 0.8686320 0.7909859 0.9329321
0.7716948
##  [197] 0.8502715 0.8908670 1.1475067 1.0671311 0.9745383 1.0073927
```

```
0.8874678
##  [204] 1.0623773 0.9593141 0.9410767 0.9402278 0.9953612 1.0018693
0.8984173
##  [211] 0.7673933 0.9177921 0.9331660 0.9592497 0.9233550 0.9109142
0.8657846
##  [218] 0.8371189 1.1997496 0.9982760 1.2596228 1.1094293 0.9587219
1.1618845
##  [225] 0.8311579 0.9505046 1.2102269 1.0348414 0.7664057 0.8399587
1.0520589
##  [232] 0.9983069 1.1139479 1.0459517 1.2298468 1.0324168 1.0209751
0.9087116
##  [239] 1.1001712 0.9487625 1.0699112 0.9862624 1.0493129 0.9003958
1.1446360
##  [246] 0.7327366 0.9183268 0.8345367 0.8838106 1.1416692 0.8760962
0.9274893
##  [253] 0.9197151 0.8739263 1.1645873 1.0964112 1.0141674 1.0373413
0.8671904
##  [260] 1.2037991 1.1102733 0.9891225 1.0396684 0.7478346 0.9276369
0.8243502
##  [267] 1.0483848 1.0784355 0.8196020 0.6435139 0.8449918 0.8434033
1.1376889
##  [274] 1.0931940 0.8355953 0.9446504 1.0107015 1.1004672 1.0825498
0.8803912
##  [281] 1.1148383 0.7658045 1.1650395 0.8071839 0.9216751 0.7879822
1.0347646
##  [288] 0.9931594 0.9786562 0.8456303 1.0753041 0.9549443 0.8768470
0.9790271
##  [295] 0.9512437 1.0090553 0.8963722 1.0677075 0.7730833 0.7030574
1.0353773
##  [302] 1.0073148 0.9551268 1.1039598 1.0512381 0.9397922 1.1298806
0.9980230
##  [309] 0.8038165 0.9965239 0.8462117 1.0780739 0.9851992 0.9044195
1.0872318
##  [316] 1.1185988 0.8645992 0.9040048 1.1661860 1.2244874 0.8245771
1.0538531
##  [323] 0.8096475 1.0065176 1.1217337 0.9363156 0.7630746 1.1227853
0.9102020
##  [330] 1.0504810 1.1286700 0.8495741 0.9399326 0.9109247 0.8030193
0.7056881
##  [337] 0.9969815 0.9491856 0.7963832 0.9483114 1.0674882 0.9958003
0.9741195
##  [344] 0.9679985 1.0690359 0.9787183 0.7728853 0.5894895 0.9333693
1.0842927
##  [351] 1.0188193 0.8345744 0.6784259 0.8661270 1.1100438 0.9255922
0.9824966
##  [358] 1.0476429 0.7754814 1.1121418 1.0287572 0.9496679 0.8060536
0.8346197
##  [365] 0.9082714 0.9286577 1.0160974 1.1871341 1.0408925 0.8779071
1.0161434
##  [372] 0.8633521 0.8494342 1.0540352 0.8111737 0.8778540 0.9878683
```

```
0.9036609
##  [379] 0.8771925 1.0909046 0.8986656 1.0345547 0.7580630 0.8752461
1.0748239
##  [386] 0.8915372 0.8693073 0.7556820 0.9194303 0.9528398 0.9179080
0.9918571
##  [393] 0.9469818 0.9222605 0.8353342 0.9543475 0.9143277 0.7988030
0.8229422
##  [400] 1.0041790 0.8966792 0.9952749 1.0136736 0.9513469 1.1078721
0.7991733
##  [407] 0.8907532 0.9910068 1.0564795 1.1911637 1.0178301 1.1040895
0.9739830
##  [414] 0.9140624 0.8438129 0.9863337 0.8939198 0.9705782 1.2885807
1.3214273
##  [421] 1.0550004 1.2033212 1.0307564 0.9121097 0.7604658 1.1415878
0.8322735
##  [428] 0.7056745 1.0177417 0.9075746 1.2744340 0.8925297 0.9375638
0.7487907
##  [435] 1.4760077 1.1083564 0.9683227 1.0487036 1.0623702 0.9978160
0.7456720
##  [442] 0.7122337 0.7854537 0.7425151 1.0224450 0.7703527 0.9054747
0.8053234
##  [449] 0.8955399 0.9563440 1.0888109 0.8895100 0.8622147 0.7438467
0.9037129
##  [456] 0.9363023 0.7817102 0.6740603 0.7540881 1.0683877 0.9053403
0.8093094
##  [463] 0.8542809 1.2809875 0.8475208 1.2059314 0.8413130 1.0053594
0.9892967
##  [470] 0.9196343 0.7393179 1.0279381 0.7659777 0.8403848 0.9835555
0.9651764
##  [477] 0.9409488 0.9527642 1.0440132 0.8121112 1.0534595 0.6355414
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##  [750] 0.9507255 1.1050234 1.1477582 0.9530502 0.9981133 1.1752710
0.8512130
##  [757] 0.9959186 0.7442756 0.8811635 0.9586478 0.6425220 0.7943146
1.1117287
##  [764] 1.0884588 1.0371396 0.8318934 1.1228528 1.1188722 1.0609649
0.7689137
```

```
##  [771] 0.9662601 1.1122419 0.8788057 0.8194633 1.0168922 1.0565055
1.1784895
##  [778] 1.1029036 0.8971518 1.1097438 0.9304455 0.9394492 0.9609774
1.0765664
##  [785] 1.0120195 0.8365487 1.0738960 0.8432060 0.9469556 0.7977836
1.1962084
##  [792] 1.0587447 0.9347311 0.8944270 1.2116215 1.0094990 0.9138958
1.0196629
##  [799] 0.8898941 0.7847512 0.9966382 0.9398703 0.7667340 1.0872260
0.8618611
##  [806] 0.7056496 0.7220644 0.9468175 0.9505328 0.9703546 0.8700967
0.9239650
##  [813] 1.0689545 1.0492786 1.1195216 0.8643914 0.8657829 0.9742803
0.8538210
##  [820] 0.9226265 0.8512042 1.1275235 0.8432064 1.2395092 1.2518125
1.2935528
##  [827] 0.9428973 1.0268709 1.0453898 1.1438628 1.2208504 1.0040346
0.9900904
##  [834] 0.7587356 0.8251138 1.0075136 1.0766458 0.8867657 1.2863825
1.0604600
##  [841] 0.8518753 0.9837611 0.9344463 0.8677111 0.9661369 0.9866563
0.8306484
##  [848] 0.8499370 0.9322052 0.9133004 1.0534351 0.9820143 1.0999641
0.9869359
##  [855] 0.9405826 0.8813308 0.8610068 0.7905097 1.0185432 0.9255889
1.1362857
##  [862] 1.0387987 0.8823332 0.8504344 0.8841840 0.9828191 0.7401638
1.1662430
##  [869] 0.9627562 1.1404850 1.0416294 0.9778481 0.7465372 1.0968608
0.8677945
##  [876] 1.0264724 1.0992874 0.9946758 0.6630985 0.7779015 1.0475250
0.8938373
##  [883] 1.1039543 1.1635989 0.9102414 0.9209075 0.8893095 1.0989529
0.7073901
##  [890] 1.1590700 0.8811212 0.8265946 1.1485188 0.8951863 0.9735569
1.0601792
##  [897] 0.8375011 1.0165636 0.8343898 0.9551163 0.8513007 0.9113405
0.8020737
##  [904] 0.9048276 1.1517662 0.9660278 1.0653617 1.0026553 0.9618279
0.8844522
##  [911] 0.8783917 0.9497463 0.8009409 0.9672396 0.8860847 0.8280819
1.2453204
##  [918] 1.1666692 1.3020120 0.8180182 0.9826560 0.8799992 1.2178025
0.6462672
##  [925] 1.0967900 0.8508074 1.0252572 0.8483772 1.0231700 1.0629312
1.1981370
##  [932] 1.0855794 0.9873546 0.8379503 1.1403405 0.8132239 0.8586995
0.8756737
##  [939] 1.1881970 0.9287282 1.0742641 0.7546869 0.7497437 1.1881718
0.9402285
```

```
##  [946] 1.0210706 1.1023214 1.0061317 1.1413350 1.0424490 0.8307539
1.0437293
##  [953] 0.8492170 1.0135811 0.9388490 1.2586681 1.0959099 1.0037027
1.1159163
##  [960] 0.9580994 1.2679540 1.1334560 0.9456961 0.8753061 0.6554439
1.1421773
##  [967] 0.9041792 0.8394908 0.6419015 1.1870238 0.8556206 0.8924291
1.2116238
##  [974] 1.0764129 0.7481794 1.1647019 0.9982562 0.9593189 0.9882880
1.0499527
##  [981] 0.8619751 1.1645632 0.8833765 1.2746876 0.9541308 0.8337491
1.1466968
##  [988] 0.7394465 1.1590651 0.9778340 1.0681420 1.2036478 0.9341402
0.8994253
##  [995] 0.8602949 0.9469253 1.0583382 1.0874459 1.0171502 1.0154366
```

When the true interaction coefficient is negligible ($\beta_3 = 0.001$), omitting the interaction term has virtually **no impact** on model performance.

```r
run_simulation(100,1000,c(-2.5,1.2,2.3,3.1))
```

```
## $avg_mse_correct
## [1] 0.9577982
##
## $avg_mse_naive
## [1] 2.863335
##
## $mse_correct
##    [1] 0.9361545 0.8990821 0.8566232 0.9398601 0.7623377 1.0244665
1.0927969
##    [8] 0.8856149 0.7915997 0.9207781 0.8624195 0.7888115 1.0326013
0.9159760
##   [15] 1.0669444 1.1187375 1.2615162 0.9316154 0.8317768 1.0679249
0.8696638
##   [22] 0.8524500 0.9713539 1.0152585 0.9934668 0.8122017 0.9048086
0.9264103
##   [29] 0.9954714 0.8660562 1.0785476 0.9752150 0.9175464 1.0012129
1.0334925
##   [36] 1.0672425 1.0688819 0.7946862 1.2908628 0.8706567 1.0668883
1.0030914
##   [43] 1.2888980 0.7544823 0.7214420 0.8893845 1.0886203 0.9427252
0.9291164
##   [50] 1.1003089 1.0845886 1.1180871 1.2392893 1.0029177 1.0108469
0.8070356
##   [57] 0.8064031 0.7093645 0.8566933 1.0079025 0.7409510 0.9532615
0.9182601
##   [64] 0.8500456 0.9764679 0.9320910 1.0800195 0.6774569 1.0899296
0.9725546
##   [71] 0.9321122 1.1354495 0.9044048 0.7843419 1.0901142 0.8032159
1.0948214
```

```
##   [78] 1.2040666 0.7739376 0.8875357 0.9301022 1.1106742 0.7988840
0.8212890
##   [85] 1.1118360 0.8559778 0.9329654 0.9083649 1.0091308 1.0682698
0.9438619
##   [92] 1.0816142 1.0258453 0.9546682 0.8410744 1.0324974 1.0265723
1.0486084
##   [99] 1.0467047 0.7269147 0.9910388 0.8849950 0.9231992 1.0574773
0.8270629
##  [106] 0.7196046 0.7609113 0.8842548 1.0306970 0.8790769 1.0020971
0.9460150
##  [113] 0.8853201 1.1917357 0.9268762 0.9011623 0.8879754 0.9437960
0.7581105
##  [120] 0.9738525 1.1062614 1.2150910 0.9586216 0.8568369 0.8471034
1.0183479
##  [127] 0.7577515 1.2114773 0.9677322 0.7423618 1.0019370 0.9230742
0.9643771
##  [134] 0.7915705 0.9821467 0.9777995 1.5013810 1.0998828 0.7930936
0.8544296
##  [141] 0.7861671 0.9389253 0.9201414 0.8587984 0.9698436 0.8854372
0.8768184
##  [148] 1.3289878 1.0844564 0.9821031 1.0257839 0.9744673 0.9482338
1.1463660
##  [155] 1.0844612 0.8827037 1.2549981 0.9623703 1.0391141 0.8846125
0.9769902
##  [162] 0.7760356 1.0962084 1.2946534 1.1204317 1.0250022 0.8453335
0.9598585
##  [169] 1.0131074 0.8387880 0.9069129 0.9678524 1.0275583 0.7888375
1.1717725
##  [176] 0.9622671 0.8626203 1.1392128 0.7040935 0.8609095 0.9891224
0.9194963
##  [183] 0.8904678 0.6334799 0.8222465 0.8332287 1.1229189 1.0053920
0.8575848
##  [190] 0.8737016 1.0208760 0.9772593 0.9648910 0.8289035 0.8645470
1.0175362
##  [197] 0.8917951 1.0294155 0.7151337 1.0031777 0.7198873 0.9474143
1.0384723
##  [204] 1.0408619 0.8980446 0.9338481 0.7300061 1.1518613 1.0189672
0.7055504
##  [211] 0.9641013 0.9038130 1.2304365 0.9981491 1.0283878 0.6450458
0.7618279
##  [218] 1.2260050 0.9869282 0.8417837 0.9037629 1.0124335 1.3857188
0.8714561
##  [225] 0.7609785 1.0057915 0.9135895 0.7111046 0.8964854 0.9698883
1.0777977
##  [232] 0.7004975 0.9953234 0.6316139 1.2028539 1.1113403 1.2118171
1.0513480
##  [239] 1.1356997 1.1116658 1.0129824 1.0437220 1.1342484 0.8691220
0.7327770
##  [246] 1.1586254 0.9615696 0.7173504 0.9411977 0.7704434 0.9289765
0.9775554
```

```
##  [253] 0.7487299 0.7824630 0.8274480 1.3002326 0.8938576 0.8035643
1.0993674
##  [260] 1.2287194 0.8497772 0.9937745 0.8931644 1.0960179 1.1586650
0.8790313
##  [267] 1.0573877 0.7693386 1.1817637 0.8361020 0.9150779 0.9569142
0.8564640
##  [274] 0.9061593 1.0379216 0.9067828 1.0574433 1.0065283 1.0435883
0.9288803
##  [281] 0.7633712 1.1455051 0.9207118 1.2124945 0.8532639 1.1009438
0.8555757
##  [288] 0.9053347 0.8384516 0.6874961 1.1281134 0.8898763 0.8710059
0.8885668
##  [295] 0.9575980 1.1575850 1.3817699 0.8340211 1.1088103 0.9161409
0.7920637
##  [302] 0.9102233 1.0979247 0.9848324 1.0193055 0.7564745 0.8219685
1.3220606
##  [309] 1.0924950 0.8111728 0.6944896 1.1547901 1.0256480 1.0032320
0.9093133
##  [316] 0.8655303 0.9178649 0.9497310 0.8771125 1.0965853 0.8211081
0.8047440
##  [323] 1.1069910 1.1536223 0.9691837 1.3330681 0.9347032 1.0971730
0.8734772
##  [330] 1.1305873 0.9444613 1.2294365 0.9882098 0.9028136 1.1622214
1.0341283
##  [337] 0.9548900 0.9563632 0.8774541 0.8500951 0.9355155 0.8207613
0.8667338
##  [344] 0.9204586 0.8253232 0.7688911 1.0182378 0.7977529 1.1179595
0.8381644
##  [351] 1.0759129 0.8597196 0.9946024 0.8572492 1.1186971 1.0283072
0.8634575
##  [358] 0.9363954 0.9951528 1.0015006 0.8601266 0.9451882 1.0068546
1.1195834
##  [365] 0.9546632 0.9382029 0.9872500 1.1233052 0.8810696 0.9729866
0.9518423
##  [372] 1.0702567 1.0911178 0.8294476 0.9627929 1.0745020 0.8016776
0.9878672
##  [379] 1.0648439 1.0362648 0.9220304 0.9232978 0.9889680 0.8337160
1.0981203
##  [386] 0.8182776 1.0595772 1.1286169 0.7667520 1.1457544 0.8718021
0.8702621
##  [393] 0.7159424 0.9295490 1.0892610 0.8987110 1.1618812 0.9114219
1.0358046
##  [400] 0.7865745 0.9409889 0.9977586 1.2079069 0.9390894 0.9861883
0.9464881
##  [407] 0.8432442 0.9151957 0.7813678 0.7431874 1.1154252 0.9265377
0.9542373
##  [414] 0.8213268 1.0288713 0.7761481 1.0237507 0.9829092 1.0500058
1.2115333
##  [421] 1.2908454 0.7736512 0.8380551 0.8810190 1.0659142 1.0887672
1.0577107
```

```
##  [428] 0.8355960 0.9189144 1.0750378 0.9011302 1.4429013 0.8436319
1.3172685
##  [435] 0.9624655 1.0289159 0.9979174 0.6698347 1.0045354 0.8387588
1.0983243
##  [442] 0.8512539 0.7609243 0.8932473 1.0240899 0.9295669 1.0871644
0.9270119
##  [449] 0.9512467 0.9370781 1.0093900 1.0227971 0.9494176 0.8998242
1.0101323
##  [456] 0.9554314 0.7990500 0.8447479 0.8493617 1.1020451 0.8341017
0.9236504
##  [463] 0.8858368 0.9675076 1.0037023 0.8917944 1.0889579 0.7324787
1.2385317
##  [470] 0.8670215 0.9592385 0.9703518 0.8358067 0.9829378 0.8363364
0.8407570
##  [477] 0.8287115 0.8359400 0.9866670 0.8592042 0.9624931 0.9105252
1.0587821
##  [484] 1.1070618 0.8802474 0.8520995 0.9041884 0.8529144 1.0392183
1.1629794
##  [491] 0.7494924 0.6825400 0.9755624 1.1127720 0.8601300 0.8815698
1.1421880
##  [498] 0.6956854 0.8787940 0.8479427 0.8118096 1.0781806 0.9081502
0.8314968
##  [505] 0.8672841 0.7164352 0.8966362 0.8892416 0.8948141 1.0261927
0.9103856
##  [512] 0.8465939 0.8379965 1.2658096 1.0028633 1.2195917 0.8407961
0.8869090
##  [519] 1.0603719 1.4216655 0.8894573 0.8657812 1.0826543 1.0480675
0.8071479
##  [526] 0.8743265 1.2236652 0.7893552 1.0776586 0.6639637 1.0025692
0.8022293
##  [533] 0.9269251 1.3137927 1.0959525 1.2861333 0.8707570 0.6367548
1.0892180
##  [540] 0.7430232 0.9556533 0.9385005 0.8484656 0.7992979 1.0299930
0.8366316
##  [547] 0.9361973 1.0656198 0.8376240 0.9852842 0.7550589 0.9076819
0.8487445
##  [554] 1.0549145 0.9921830 0.9781893 0.7408467 1.0327859 1.0724908
0.8807847
##  [561] 0.9115610 0.8247370 0.9281429 1.0536435 0.7529272 0.8750468
0.9607401
##  [568] 0.8742492 0.8174630 1.0642960 1.1835942 1.0146097 1.0383338
0.8359485
##  [575] 0.8912720 0.9239016 0.8194240 1.1507481 0.8668135 0.7519268
0.9594883
##  [582] 0.9234982 0.9280818 0.6334354 0.8366060 0.8830839 1.1140614
0.9531276
##  [589] 0.9674081 0.8180888 1.0416375 1.1500163 1.0683057 0.9825940
1.2637607
##  [596] 0.8315517 0.8029067 0.8350900 0.9704504 0.8986476 1.1467176
0.8119457
```

```
##  [603] 0.7920981 1.3639971 0.9690239 1.1068627 0.8631864 0.9205244
0.8141214
##  [610] 0.8752052 0.9066993 1.3770309 0.9360817 1.1316953 1.0139916
0.9124212
##  [617] 1.0777083 0.8949408 1.0115264 1.0257097 1.0783497 0.8539070
0.8600946
##  [624] 0.9023828 0.9290858 1.1350814 0.8525150 0.7715496 0.9948620
0.9324538
##  [631] 1.0643536 1.1786501 1.1574553 1.1945426 0.8092926 1.0622068
0.7528540
##  [638] 0.8231873 0.9636972 1.0404842 0.9307342 1.0117032 0.9305865
0.9800333
##  [645] 0.9173899 1.1045748 0.6658585 0.8940871 0.9309087 0.8931127
0.8575329
##  [652] 0.8253317 1.1762863 1.0503267 0.9184740 1.2556025 0.8538276
0.6591595
##  [659] 0.8195373 1.0567908 1.1310585 1.0904853 0.8527602 0.9973798
0.8948228
##  [666] 1.1493733 0.8909250 0.9420281 0.8324161 1.2781602 0.9529982
0.9075431
##  [673] 0.8220712 0.9164201 1.2011484 1.0312551 1.1055412 0.9690627
0.5821412
##  [680] 0.8935384 0.8637832 1.2702192 0.9947389 0.7882127 0.9953438
0.9111276
##  [687] 0.9255896 1.0174543 0.8420265 0.8427056 0.9926572 0.8562309
0.8762321
##  [694] 1.2671003 0.8936975 1.2632574 1.1090707 0.8814327 0.8103761
0.9135801
##  [701] 0.7732975 0.9953630 1.0723614 1.3234064 0.9519407 0.9517217
0.9083417
##  [708] 0.7225536 0.9629567 0.7825433 0.7924804 0.8137483 0.6907267
1.0522900
##  [715] 0.7641276 0.8483272 0.9612704 1.0976159 1.0656334 0.7315605
0.9731643
##  [722] 0.9481463 0.8240111 0.8161603 1.0579173 0.8676753 1.0813281
1.2033264
##  [729] 0.9231347 1.1567936 0.9718757 0.7265016 1.1422417 0.8588174
0.9202780
##  [736] 0.7457474 0.8644985 0.9841914 1.1606256 1.0709791 0.9511585
0.9627881
##  [743] 0.9837385 1.0629350 0.9398662 0.8445375 0.9132849 1.0080695
1.1335591
##  [750] 0.9320067 0.6400744 0.9494034 0.9618902 0.8350247 0.8616564
1.1902512
##  [757] 0.8946523 0.8955946 0.9081948 0.8225404 0.7837904 0.9691035
0.9038926
##  [764] 0.7525610 1.1076856 0.9945674 0.9289282 0.8470829 0.9635789
1.1550021
##  [771] 0.8439355 0.9406165 0.9910170 1.1132723 1.2365481 0.8368970
0.9958011
```

```
##  [778] 0.9444941 1.0720599 0.7915311 0.9947416 1.1925672 0.8171506
0.9736184
##  [785] 0.9037952 0.9239552 1.0883917 1.0857878 1.1288658 0.9378762
0.9761206
##  [792] 0.8730052 0.9099778 0.9186982 0.8296747 0.9569123 0.9994078
0.9263066
##  [799] 0.9553606 1.2018729 1.1719993 0.9677010 0.9188100 0.8856086
0.9156362
##  [806] 0.9106329 0.8452239 0.8218903 0.8929684 1.1576707 0.9329780
0.9380204
##  [813] 0.9972323 0.8923117 1.0782730 0.7178586 0.9596519 0.9587926
0.8699858
##  [820] 1.1287530 1.1039863 0.9184469 0.7638602 0.8988526 1.0126249
0.9570140
##  [827] 1.0283982 0.7838669 0.8524976 1.0236651 0.9911209 1.1424525
1.0556602
##  [834] 1.0509653 1.0340520 0.8618209 0.8310007 1.1546950 1.1073725
0.7848384
##  [841] 0.9890584 0.7965154 0.9613502 1.2201562 1.0925344 0.9953520
1.1026482
##  [848] 0.8073789 1.1913812 0.9311888 1.1323489 0.9778743 1.2981842
0.9530511
##  [855] 0.8707661 0.8916743 0.9833332 0.9153925 1.0541287 0.8617953
0.9308750
##  [862] 0.8139517 1.0000136 0.9684914 1.0778692 1.1296463 0.9222533
1.0779847
##  [869] 1.0843978 0.7222866 1.0238751 0.9131255 0.7969416 1.0346804
1.0299325
##  [876] 1.0712069 0.9494309 0.7457409 1.0002841 0.9634823 0.9898179
1.1179486
##  [883] 1.3907815 1.0810953 1.0505139 1.0013521 1.2141436 0.9265848
0.8713050
##  [890] 0.8585403 0.9895619 0.8276423 0.9684373 0.8887241 1.0186876
0.7186916
##  [897] 1.0475940 0.9832025 0.8033251 0.7241361 0.8724473 1.2434902
0.9633109
##  [904] 0.8595720 0.7845791 0.8061616 0.9163650 1.1045607 1.1447460
0.9332677
##  [911] 0.9459503 0.8572492 0.9661778 1.0026917 1.0829949 1.2730644
0.9212672
##  [918] 1.0408293 0.9639702 1.0258519 0.7952719 0.8330229 1.0525655
1.0237804
##  [925] 0.9066035 1.0666364 1.0529176 0.9669374 1.2124933 1.0470891
0.9267243
##  [932] 1.1676620 0.6517698 1.0148729 1.1072961 1.1185751 0.9160450
0.9311672
##  [939] 0.9532218 0.8904106 0.9768558 1.2630619 0.8387899 0.7187059
0.8011383
##  [946] 1.0703580 1.1291891 0.8649717 0.8714955 1.0675351 0.7733437
1.2366335
```

```
##  [953] 0.8942361 1.0012290 0.7202741 0.9266358 0.8627683 0.9032598
1.0498782
##  [960] 0.8945491 0.7811256 0.7567961 1.0253583 1.0020615 1.0023253
0.8469759
##  [967] 1.0454423 0.9802831 0.7864159 0.9920537 1.1531105 0.9415514
0.9337490
##  [974] 0.8497009 1.0446896 1.1374754 0.9750601 0.7651750 0.7802087
1.1298226
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When the true interaction coefficient is large ($\beta_3 = 3.1$), omitting the interaction term has a **big impact** on model performance.